

Yiming Gao

Ph.D. Student in Cyberspace Security, University of Science and Technology of China

qw1234567@mail.ustc.edu.cn | Hefei, China

RESEARCH INTERESTS

Lattice algorithms and discrete Gaussian sampling; algorithmic number theory and provable cryptanalysis; subset-sum and related computational problems.

EDUCATION

University of Science and Technology of China (USTC)

Ph.D. in Cyberspace Security; Advisor: Prof. Honggang Hu

Hefei, China

Sep. 2023 – Present

University of Science and Technology of China (USTC)

B.Eng. in Information Security, GPA: 3.47/4.30

Hefei, China

Sep. 2019 – Jun. 2023

PEER-REVIEWED PUBLICATIONS

1. **Better Bounds for Finding Fixed-Degree Isogenies via Coppersmith’s Method**
Marius A. Aardal, Diego F. Aranha, Yansong Feng, **Yiming Gao**, and Yanbin Pan.
EUROCRYPT 2026. [DOI] *Authors are listed alphabetically.*
2. **On Factoring and Power Divisor Problems via Rank-3 Lattices and the Second Vector**
Yiming Gao, Yansong Feng, Honggang Hu, and Yanbin Pan.
Mathematics of Computation, 2026. [DOI]
3. **Attacking ECDSA with Nonce Leakage by Lattice Sieving: Bridging the Gap with Fourier Analysis-Based Attacks**
Yiming Gao[†], Jinghui Wang[†], Honggang Hu, and Binang He.
ASIACRYPT 2024. [DOI] [†]*Equal contribution.*

PREPRINTS

1. **Provable Subexponential Algorithms for NIST Third-Round Lattice Families**
Yiming Gao, Xuyuan Han, and Honggang Hu.
Preprint, 2026.
2. **A Provable Subexponential-Time Algorithm for LWE from $n + o(n)$ Samples via Wagner-Style Gaussian Sampling**
Xuyuan Han, **Yiming Gao**, and Honggang Hu.
IACR ePrint, 2026. [ePrint]
3. **Provable Recovery of RSA Private Exponents below $N^{11/42-\varepsilon}$**
Yiming Gao and Honggang Hu.
IACR ePrint, 2026. [ePrint]
4. **Discrete Gaussian Sampling Meets BDGL Decoding: Solving the Shortest Vector Problem in $2^{0.5596n+o(n)}$ Time**
Yiming Gao, Yansong Feng, and Honggang Hu.
IACR ePrint, 2026. [ePrint]
5. **Solving the Shortest Vector Problem in $2^{0.7314n+o(n)}$ Time via Discrete Gaussian Sampling on Superlattices**
Yiming Gao, Yansong Feng, and Honggang Hu.
IACR ePrint, 2026. [ePrint]
6. **Faster SVP in Polynomial Space**
Yansong Feng, **Yiming Gao**, and Jiaqi Liu.
IACR ePrint, 2026. [ePrint]
7. **Subset Balancing and Generalized Subset Sum via Lattices**
Yiming Gao, Yansong Feng, Honggang Hu, and Yanbin Pan.
arXiv preprint, 2026. [arXiv]

SELECTED TALKS

- Rank-3 Lattices and the Second Vector in Deterministic Factoring** Sep. 2026
Invited talk, Academy of Mathematics and Systems Science, Chinese Academy of Sciences (online).
- Better Bounds for Finding Fixed-Degree Isogenies via Coppersmith's Method** May 2026
Conference talk, EUROCRYPT 2026.
- Attacking ECDSA with Nonce Leakage by Lattice Sieving: Bridging the Gap with Fourier Analysis-Based Attacks**
Invited talk, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China. Nov. 2025
Conference talk, ASIACRYPT 2024. Dec. 2024

ACADEMIC SERVICE

External Reviewer: 67th IEEE Symposium on Foundations of Computer Science (FOCS 2026)

TECHNICAL SKILLS

Programming: C, Python, SageMath

Tools: Linux, shell scripting, L^AT_EX